Telemedicine: Trend Analysis

Sehhaty Application

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1. Executive Summary:

In recent years, a relatively new field has emerged in healthcare: Telemedicine. Telemedicine is the practice of providing patients with services including diagnosis, examination, and medical assessment via means of information technology and electronic communication. [1]. With the rapid progress in communication and Internet technologies, telemedicine has witnessed tremendous development, as many diseases and health conditions have become capable of being diagnosed and treated remotely.

This study describes Sehhaty application. Sehhaty application is an electronic application provided by the Ministry of Health in the Kingdom of Saudi Arabia by providing health information and related services to more than twenty-four million beneficiaries. The Sehhaty application helps many people save their time and effort. This study purpose to introduce this application and the reasons for its fame and spread among users from all segments of society.

1.1. Objectives Of the Case Study:

- Services of the application.
- User interfaces of the application.
- Reasons for the spread of the Sehhaty application.

2. FINDINGS:

2.1. Services of the application:

- Ability to book a virtual appointment: This service provides a free appointment booking for a remote medical consultation.
- Booking an in-person appointment: This service provides the ability to book an appointment at the hospital.
- Health Record: This service provides the ability to obtain all your health data.
- Medication: This service provides ability to choose the medication you may need.
- Prescriptions: This service provides easy access to medical prescriptions.
- Vaccines: This service provides detailed information about all vaccinations previously taken.
- Visits: This service provides the number of visits you have made previously with all their details.

2.2. User interfaces of the application:

When creating an application, one of the most important factors is the user interface, or how the application appears to the user. [2]. It is the part of the application that the user interacts with. It includes all the visual and interactive sections that the user controls in the application and performs basic tasks. The goal of the user interface is to make interaction with the application easy and smooth, while providing an easier and less user experience.

Sehhaty application contains four main interfaces:

- Home.
- Appointments.
- Well-being.
- My File.

2.2.1. Structure and Navigation:

- Overall Structure: The Sehhaty application features a logically organized user interface.
- Navigation Bar: The navigation bar contains the four main interfaces that were mentioned above.

2.2.2. Visual Design:

- Color and Typography: Sehhaty application uses consistent colors and easy-to-read fonts.
- Icons and Symbols: The Sehhaty application has intuitive icons and symbols that clearly represent functions.

2.2.3. Usability:

- Ease of learning: The Sehhaty application is characterized by an easy-to-understand and use interface.
- Simplicity of interaction: The Sehhaty application features smooth navigation.
- Fast Response: Sehhaty application features a fast response interface.

2.2.4. Feedback:

- Confirmation messages: The Sehhaty application features a message appearing immediately upon completion of any process.

2.3. Reasons for the spread of the Sehhaty application:

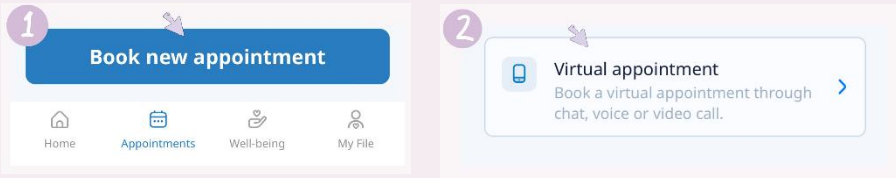
The Sehhaty application has spread widely in the Kingdom of Saudi Arabia recently, and the reasons for its spread can be summarized in the following points:

- **Technological development and communications:** The increased use of smartphones contributed to the rapid spread of the Sehhaty application.
- **Access to remote and rural areas:** Ability to provide consultations and request treatment without having to go to the hospital.
- **Growing demand for effective healthcare:** Sehhaty app helps organize appointments and reduce pressure on hospitals.
- **Responding to global health crises:** The COVID-19 pandemic has accelerated the spread of the Sehhaty application.
- **Increased health awareness:** The Sehhaty application has contributed to increasing awareness of the importance of health in general.

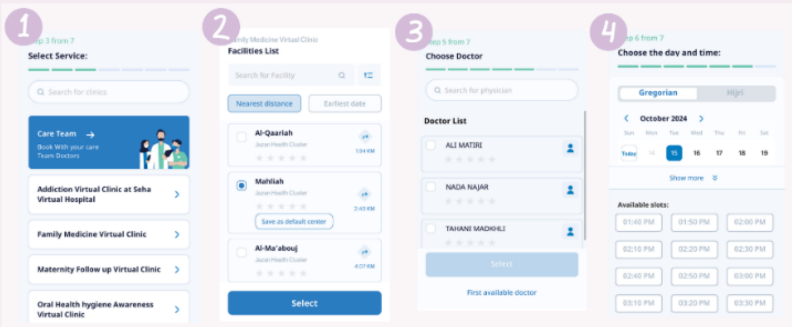
3. Discussion:

3.1. Services of the application:

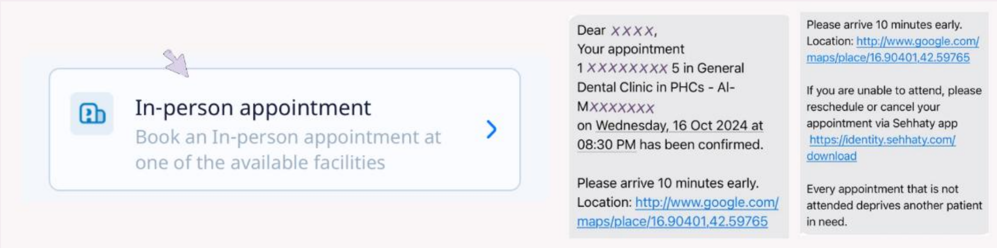
3.1.1. Ability to book a virtual appointment: The remote appointment is set in the Sehhaty application by first going to the "appointments" icon in the taskbar, secondly clicking on the new appointment booking icon, and thirdly clicking on the virtual remote appointment icon.



The last four steps include: Determining the type of service that suits you according to your needs, for example: family medicine clinic. Then choosing a facility that suits you according to the nearest location or appointment time. After that, we choose a doctor from the list of doctors. Finally, the date and time are chosen.

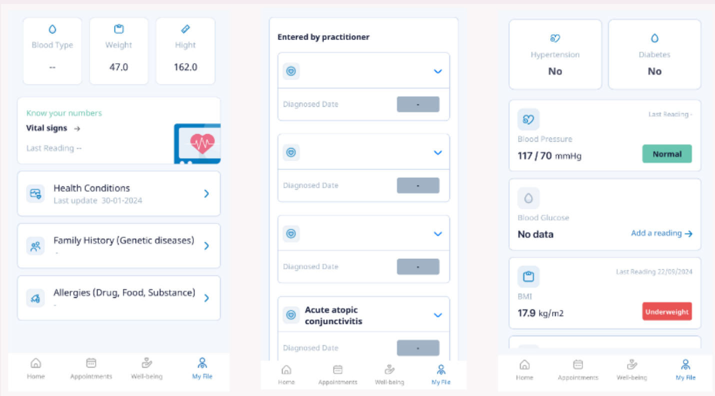


3.1.2. **Booking an in-person appointment:** You can also book an in-person appointment through the Sehhaty application by following the same steps mentioned in the **"Ability to Book a virtual Appointment"** section, except that you must choose the in-person appointment icon and then complete the steps explained above.

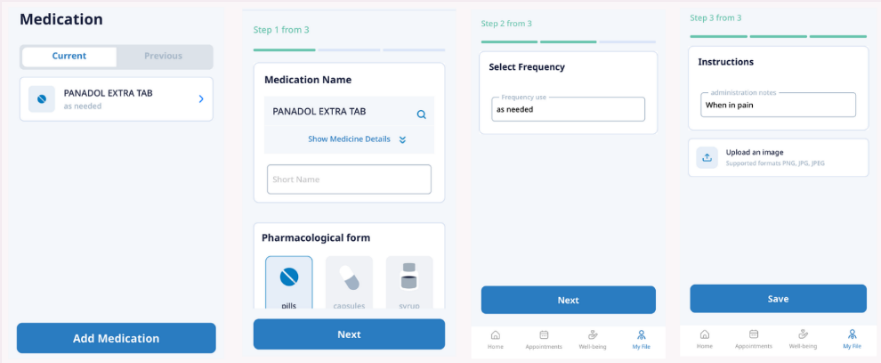


One of the features available in the Sehhaty application for this service is that it reminds you one day before the appointment by sending a message that includes: the location, the name of the service, and the date and time of the appointment. In the event of any circumstance for the patient, the appointment can be rescheduled or cancelled.

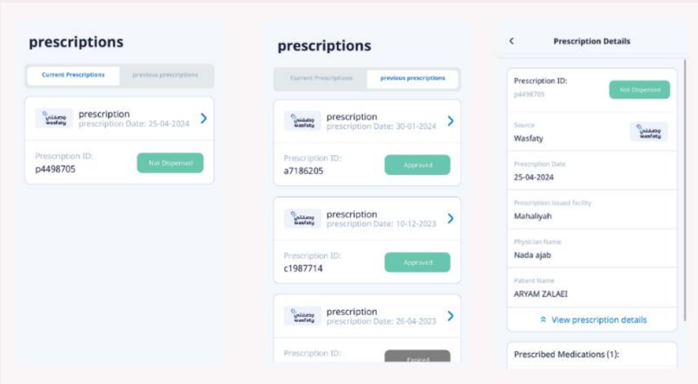
3.1.3. **Health Record:** This service provides comprehensive health information starting from your blood type, height and weight, to your family medical history, and allergies to certain medications, foods, substances, etc. This service also includes vital signs (blood pressure, blood sugar, body mass index, heart rate, body temperature, etc.), and health conditions whether entered by a doctor or the user. The fact that the information is collected in one place contributes to its easy display and retrieval.



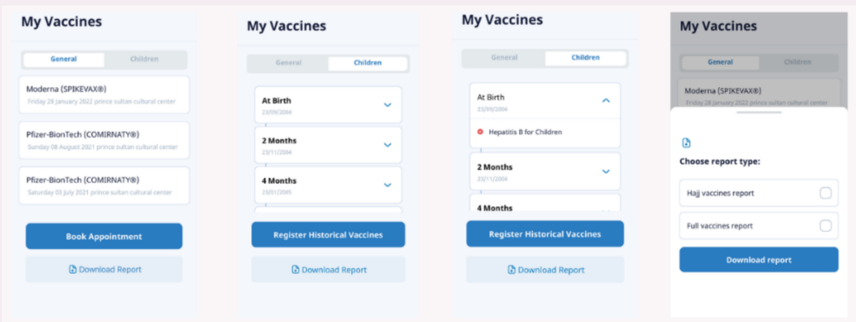
3.1.4. **Medications:** This service provides the ability to review previous and current medications so that you can view the details of this medication from dosage data (drug form, medication frequency, and medication usage notes) and its general information (generic name of the medication, medication dosage unit, storage method, etc.). This service also provides the ability to add a medication by the user by writing the medication name and choosing the medication form. Secondly, the number of times the medication is taken is determined. Finally, the medication usage notes are determined.



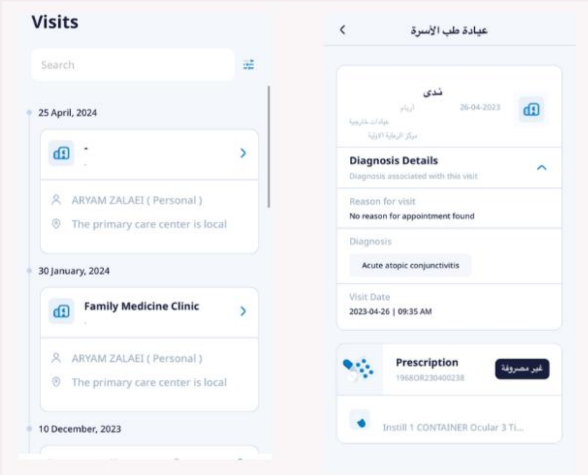
3.1.5. Prescriptions: This service collects all current and previous prescriptions. Each prescription contains the following data (prescription number, source and date of issue, name of physician and references, etc.) and the prescribed medications.



3.1.6. Vaccines: This service includes general vaccinations (Pfizer) and children's vaccinations (starting from birth from the age of four to six years). Under each age group are the names of the vaccinations that have been taken and those that have not been taken. This service also provides the ability to add a report, whether it is a report for Hajj vaccinations or a complete report for vaccinations. You can also book an appointment as explained in "Book a virtual appointment"



3.1.7. **Visits:** This service contains all the visits you have previously made, where each visit contains: the doctor's name, the patient's name, the type of clinic, the location of the clinic, the diagnosis details (reason for the visit, the diagnosis, the date of the visit), and the medical prescriptions.

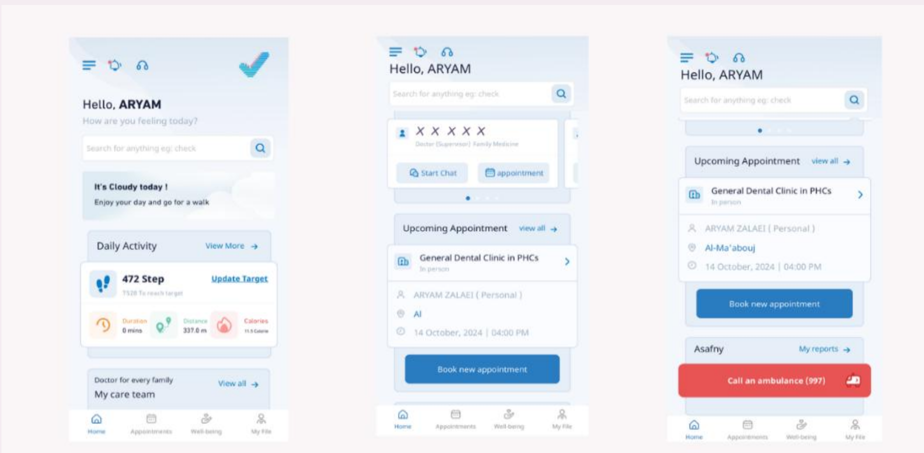


3.2. User interfaces of the application:

Sehhaty application contains four main interfaces:

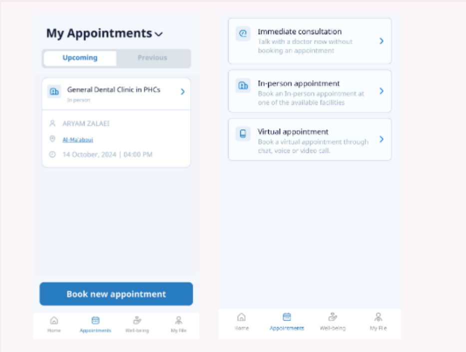
- **Home**:

This user interface contains a variety of icons and more, which are: a welcome phrase for the user, an icon for customer service, an icon that collects alerts for the user (to remind him of appointments, for example), and there is also a list containing (settings, logging out, changing the language, and an overview of the Sehhaty application, etc.), the daily activity box will be detailed in the **well-being** interface, the medical team box (contains the names of the members of the medical team assigned to the user's family), the upcoming appointments box will be indicated in the **appointments** interface, the "**Asafny**" icon where you can file an emergency report (by specifying the number of injured, specifying the location and type of report) and reviewing current and previous reports**.**



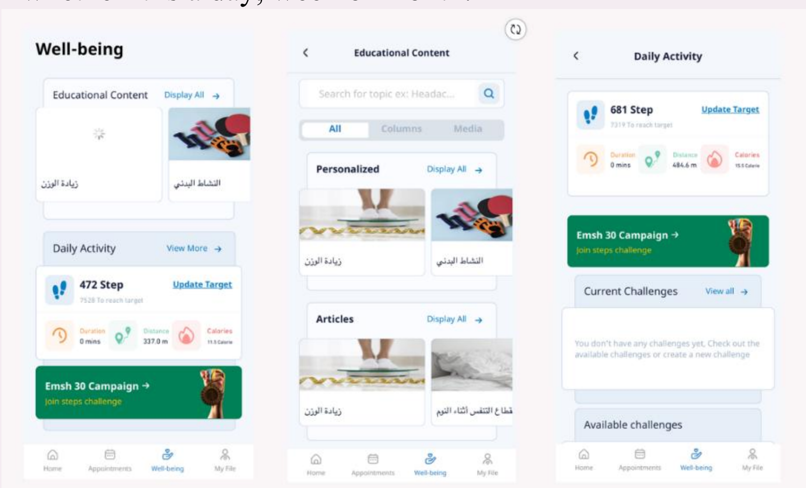
- **Appointments:**

This interface contains current and previous appointments with the ability to specify the user. At the bottom there is an icon for booking a new appointment, which contains an immediate consultation (enabling you to talk to the doctor without booking any appointment in advance), a virtual appointment and an in-person appointment, which were explained previously.



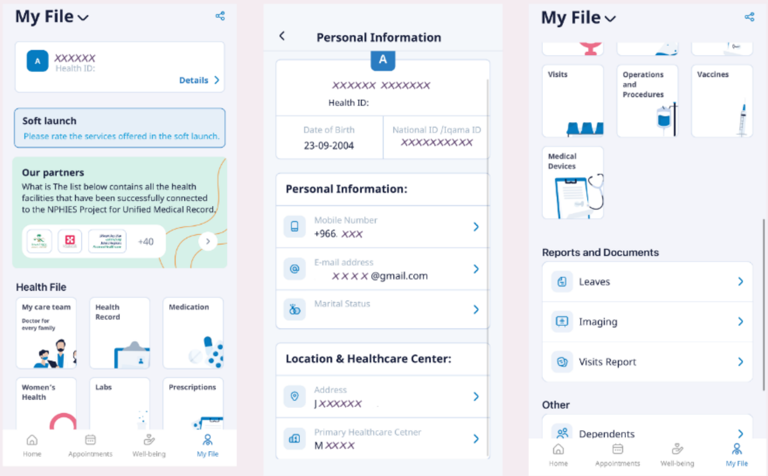
- **Well-being:**

The Well-being interface contains a box for educational content (where you can search for a specific topic or use the following divisions: All [contains all the boxes in "Texts" and "Media"], Texts [contains the following boxes: Articles, FAQs, Tips], Media [contains posts, events, videos]), a box for daily activity (displays your daily steps [time, distance, calories] with the ability to update the goal) and contains the following boxes: Current challenges, available challenges, achievements, with a graph of the step rate whether it is a day, week or month.



- **My File:**

It contains a field for the user's personal data (username, health number, and other data). The health file contains a set of icons for services, which are (the medical team, health record, medications, women's health, laboratory tests, prescriptions, visits, operations, procedures, vaccinations, and finally the medical devices that were previously detailed). It contains reports and documents (vacations, x-rays, visit summary), dependents, cards, and insurance approvals.



3.2.1. Structure and Navigation:

- **Overall Structure:** In apps like Sehhaty, it refers to how content and services are organized and distributed within the app. The goal of the overall structure is to make using the app simple and logical for users, while making key features and health services easily accessible. This involves breaking down information and tools into clear and organized chunks, ensuring that each element is in the right place, easy to access and understand.
- **Navigation Bar:** It is an essential design element that helps the user quickly navigate between different parts of the app. The navigation bar acts as a visual guide and provides direct access to important features or pages in the app without having to return to the home screen. In an app like Sehhaty, it is often placed at the bottom of the screen (for easy use on mobile phones)

3.2.2. Visual Design:

- **Color and Typography:**
  - ➢ **Color:** The “Sehhaty” app relies on calm and formal colors associated with health and professionalism. The main colors in the app are

usually inspired by shades of blue, which is a color that expresses calm, trust, and cleanliness. Blue is associated with trust and safety. The white background is widely used in the app to help highlight texts and icons and provides a sense of simplicity and clarity. White is the ideal color to give a sense of purity and clarity, which is suitable for health apps. The app contains some complementary colors such as shades of gray and light blue used to divide sections or distinguish content, while red is used for notifications or warnings (such as alerts about appointments, calling an ambulance, or important medical notices).

- **Typography:** A simple and easy-to-read font is often used in Arabic and English. The app also relies on different font sizes to highlight importance; titles are larger and clearer, while smaller sizes are used for explanatory texts and notes. These differences help organize information and make the content easier to understand. Texts are also organized with appropriate spacing between lines to increase readability and avoid visual crowding.

- **Icons and Symbols:**
  - **Icons:** The icons in the "My Health" application rely on simple and easy-to-understand designs, where each icon directly expresses its function. For example:
    - Profile icon: refers to the user's health file and medical history.
    - Vaccinations icon: refers to vaccinations previously taken and other services for this service (explained earlier).

  Flat icons are used without color gradients or visual complexities, which makes them in line with modern design trends and ensures that the icons are light and fast to load. The same icons are also used in all parts of the application to maintain visual consistency and reduce the learning curve for the user. If the user sees a specific icon in one section, he will know that it performs the same function in other sections.

  - **Symbols:** such as icons that indicate health procedures or instructions, are used in a clear and easy-to-understand manner. These icons help in interacting with the application and presenting information quickly and without confusion. Icons such as the bell are also used for alerts to show important information or instructions. These icons distinguish notifications from other elements and immediately attract the user's attention. In some cases, small icons or images are used to indicate health conditions or medical procedures (such as a syringe for vaccination or a heart to measure heart rate), making it easier for the user to quickly understand the procedure or condition.

3.2.3. Usability:

- **Ease of learning:**

  The application has an intuitive design, with the page relying on a simple interface that enables new users to quickly understand how to use it. For example, in the appointments interface, clear steps are provided with explanatory text instructions such as “Choose date” and “Choose health center”, choosing the appointment, and finally confirming the booking making it easy to learn how to navigate the app instantly.

  Familiar and expressive icons are used (such as a calendar for booking appointments) which helps users understand the functions without the need for additional explanation. Each step is specific and simple without being complicated. The interaction is direct for the buttons and interactive elements are clear and large enough to avoid errors, such as the “Next” or “Confirm appointment” buttons, ensuring that interacting with the app does not require effort or extra thinking.

- **Simplicity of Interaction:**

  The app has clear steps, with tasks on the page being broken down into easy steps. For example, in the appointments interface, the user is guided through choosing the type of service, then choosing the appointment, and finally confirming the booking. Each step is specific and simple without being complicated. The interaction is direct for the buttons and interactive elements are clear and large enough to avoid errors, such as the “Next” or “Confirm appointment” buttons, ensuring that interacting with the app does not require effort or extra thinking.

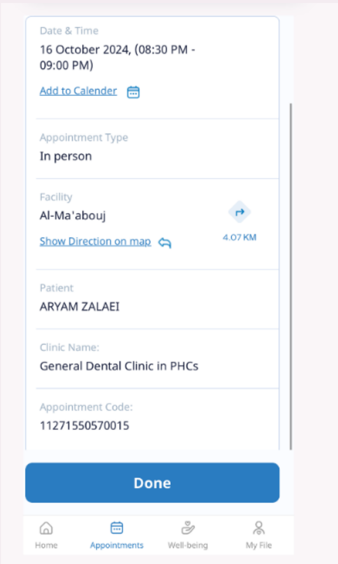
- **Fast Response:**

  The application’s immediate response is designed to be responsive, with results displayed or the next page navigated immediately after interact. For example, once a date or health center is selected on the appointment booking page, the next options are loaded immediately. The application also loads data quickly, whether the user is trying to book an appointment or access their health file, the data is displayed quickly, reducing waiting time and making the interaction more efficient and smoother.

3.2.4. Feedback:

- **Confirmation messages:**

  Confirmation messages in the “Sehhaty” app play an important role in providing immediate feedback to the user after completing various processes, such as booking an appointment or sending an inquiry. These messages appear clearly and with a simple design, to confirm the success of the process or guide the user on what to do next. For example, after booking an appointment, an immediate confirmation message appears with the necessary details (date, time, appointment type, facility, patient and clinic name, appointment code), reassuring the user that the process was completed successfully and reducing anxiety or confusion. These messages enhance user confidence and facilitate continued interaction with the app.



3.3. Reasons for the spread of the Sehhaty application:

3.3.1. Technological development and communications:

One of the main reasons that have contributed to the widespread spread of Sehhaty application. With the rapid advancement in communication technologies, access to healthcare has become easier and more flexible. Here is a breakdown of how this development has affected the growth of telemedicine:

- **Expanding the use of smartphones:**
  - With the widespread use of smartphones around the world, patients can easily access the Sehhaty application through their mobile

devices. It provides easy-to-use interfaces that enable patients to book appointments, make video calls, and obtain medical test results through the application.

- The continuous development of smartphone technologies, such as high-resolution cameras, helps doctors closely monitor medical conditions, such as dermatitis or follow-up on the healing of wounds after surgeries.

❖ **Development of cloud computing:**

Cloud computing plays a pivotal role in providing a secure and reliable infrastructure for storing and sharing health data. Thanks to it, doctors and patients can access medical records easily and from anywhere, making it easier to follow up on health conditions around the clock

❖ **Wearable Technologies:**

Smart wearable devices such as smart watches and health monitoring devices provide accurate data on patient vital signs such as heart rate, blood oxygen level, and body activity. These devices can be linked to the Sehhaty app to provide doctors with live information that helps monitor health status and adjust treatment on an ongoing basis.

❖ **Information Security and Data Encryption:**

With the increasing reliance on technology in healthcare, data security and encryption technologies have also evolved, ensuring the protection of patient privacy and the integrity of sensitive health information. Modern encryption technologies have become essential in securing communication between doctors and patients and preventing any attempts at hacking or leaking information.

3.3.2. Access to emote and rural areas:

❖ **Improving access to healthcare:**

In rural and remote areas, residents may face significant difficulties in reaching hospitals or specialized clinics due to long distances or poor transportation. Applications such as “Sehhaty” enable individuals to communicate with doctors, obtain medical consultations, and conduct periodic check-ups remotely without having to travel long distances. This reduces geographical challenges and allows individuals to obtain comprehensive healthcare regardless of their location.

- **Overcoming the shortage of medical personnel:**
  Many remote areas suffer from a shortage of specialized medical personnel, whether doctors or nurses. The “Sehhaty” application allows these residents to access specialized doctors in major cities, where they can conduct medical consultations via video or obtain prescriptions remotely. This contributes to improving the quality of healthcare and reduces reliance on limited rural clinics that may lack the required expertise.
- **Direct interaction with the healthcare system:**
  Through “Sehhaty”, citizens in remote areas can interact directly with the national healthcare system, book medical examination appointments, order medications, and access test and laboratory test results easily. This process makes healthcare more flexible and seamless and reduces the complications that patients may face in traditional settings. This continuous and direct interaction contributes to improving the health awareness of the population and encourages them to follow up on their health regularly.
- **Preventive healthcare and continuous follow-up:**
  In rural areas, preventive healthcare may be weak, which leads to the exacerbation of chronic diseases such as diabetes and high blood pressure. Using “Sehhaty”, residents can monitor their health status periodically and conduct basic check-ups remotely, which contributes to the early detection of diseases and avoiding serious complications. In addition, the application provides services such as following up on vaccinations for children, and health advice for pregnant mothers, which enhances the health of the community in general.

3.3.3. Growing demand for effective healthcare:

The growing demand for effective healthcare is one of the main reasons behind the spread of the “Sehhaty” app in the Kingdom of Saudi Arabia. With the increasing population and the exacerbation of chronic diseases such as diabetes and high blood pressure, it has become necessary to provide fast and effective medical services that meet the needs of patients immediately.

The “Sehhaty” app meets this demand by facilitating access to remote medical consultations, booking appointments, and obtaining medications and medical results quickly, which reduces waiting times and relieves pressure on hospitals.

The app is an effective way to improve the quality of healthcare, as it allows patients to monitor their health status continuously without the need to visit the doctor repeatedly, and enhances the use of digital technology to improve the operational efficiency of the health system, which leads to better directing medical resources.

3.3.4. **Responding to global health crises**:

Responding to global health crises was a key factor in the spread of the “Sehhaty” application in the Kingdom of Saudi Arabia. During crises such as the COVID-19 pandemic, it became necessary to find innovative ways to provide healthcare while reducing the risks associated with direct interaction between patients and doctors. The “Sehhaty” application came as an effective solution to meet this need, as it allowed patients to obtain remote medical consultations, monitor their health status, and obtain test results without the need to visit hospitals, which contributed to limiting the spread of the virus. The spread of infection has also been reduced by providing remote medical consultations The application contributed to minimizing the need for hospital and clinic visits, which, in turn, decreased the chances of infection spreading among individuals. The application enabled individuals to access healthcare services during a time when healthcare facilities were under immense pressure, helping manage the crisis by ensuring the safe continuity of care. The Sehhaty application also played a key role in facilitating vaccine distribution by organizing vaccination appointments during the Covid-19 pandemic, which accelerated the national immunization process. Additionally, it provided health awareness on virus prevention, making it a vital tool in managing health crises.

3.3.5. **Increased health awareness:**

Increasing health awareness among citizens was one of the main factors that contributed to the spread of the “Sehhaty” application in the Kingdom of Saudi Arabia. With the increasing awareness of the importance of disease prevention and regular health monitoring, individuals have become keener to access health information and services easily and quickly. The "Sehhaty" application meets this need by providing a wide range of digital health services, such as medical consultations, booking appointments, and obtaining test results and vaccinations, which enhances the empowerment of individuals in managing their health.The application also contributes to providing comprehensive awareness content on disease prevention, healthy lifestyles, and dealing with common medical conditions, which helps spread health awareness on a large scale. These features helped encourage people to use the application as a primary means of managing their health effectively.

4. Recommendation:

We would like to share with you some recommendations and additions that we believe will enhance the user experience in the Sehhaty application, which will increase its effectiveness and ease of use. Through these suggestions, we aspire to provide new features that better meet the needs of users. Starting from:

4.1. Adaptation to different screens:

- Description:

The "Sehhaty" application needs to be compatible with different sizes and types of screens (smartphones, tablets, etc.), to ensure a comfortable and smooth user experience. This requires improving the user interface (UI) and adapting the design to large screens, in line with the dimensions of each device.

- What to do:
  - Conduct a comprehensive analysis of all devices used by users.
  - Design adaptable interfaces using flexible techniques such as CSS Media Queries and Responsive Layouts.
  - Test the application on different devices to ensure its compatibility.

- **Responsible for implementation:**

User interface design team and software development team.

- **Expected time:** 3-4 weeks (including design, development, and testing).
- **Approximate cos**t: Depends on the work team, but in general, costs may range between 30,000 and 50,000 Saudi riyals

4.2. Ability to switch to dark mode:

- **Description:**

  Dark Mode has become a favorite among many users because it helps reduce eye strain and conserve battery power on smartphones. Adding this option will give users the freedom to choose between normal and dark mode.
- **What to do:**
  - ➢ Redesign the application interface to suit dark mode, focusing on appropriate colors and the required contrast
  - ➢ Adjust backgrounds and texts to ensure clarity of vision and ease of use.
  - ➢ Develop an easy switching option between the two modes (dark and normal) in the application settings**.**
- **Responsible for implementation:**

  UI design team and development team.
- **Expected time:**

  3-4 weeks (including design, development, and testing).
- **Approximate cost:**

  About 30,000 to 50,000 Saudi riyals

4.3. Adding a medicine image in the medicine box:

- **Description:**

  Instead of uploading a medicine image manually, the image will be displayed when searching for the medicine name within the application. This addition will enhance the user experience by facilitating visual identification of medicines, which helps avoid errors and provides a clear reference for users about the medicines they are taking.
- **What to do:**
  - ➢ Update the medicine database: Add updated and high-quality images of medicines available in the Saudi market, and provide them within the application.
  - ➢ Link the search to images: When the user searches for the medicine name, the medicine image appears directly next to its name for greater clarity.

- User experience tests: Ensure that the images are consistent with the correct medicines in all cases and do not affect the speed of the application.

- **Responsible for implementation:**
  - Database management team to add images and update medicines.
  - Software development team to integrate images with the current search system.
  - User experience team to test this feature and ensure its smoothness.
- **Expected time:**

  6-8 weeks (including database update, development, testing, and quality assurance).
- **Approximate cost:**

  50,000 to 80,000 Saudi riyals, depending on the number of medications listed in the database and the quality of the images required.

5. Conclusion:

The study reviews the "Sehhaty" application as a model for telemedicine in Saudi Arabia, providing users with services such as appointment booking and remote health consultations. This reduces visits to healthcare facilities, saves patients time and effort, and helps relieve pressure on hospitals. The application also enhances health awareness and offers flexible solutions for medical care, especially in rural areas, and played a significant role during the COVID-19 pandemic by providing remote consultations.

It is recommended to develop new features for the application, such as adapting to different screen sizes and adding a "night mode" to reduce eye strain. In conclusion, the study indicates that the future holds significant opportunities for telemedicine applications in enhancing healthcare. The "Sehhaty" application represents a step in the right direction towards comprehensive and flexible healthcare for all segments of society, with expectations for continued development and innovation.

6. References:

[1] "Telemedicine," Ministry of Health, [Online]. Available: https://www.moh.gov.sa/en/Ministry/Information-and-services/Pages/Telemedicine.aspx.

[2] D. G. a. S. A. U. I. R. V. U. a. i. s. a. i. a. a. t. l. Software AG, "Reston, VA, USA, and/or its subsidiaries and/or its affiliates and/or their licensors," Software AG, Darmstadt, Germany and/or Software AG USA, 1979-2017. [Online]. Available: https://documentation.softwareag.com/natural/nat827mf/pg/pg_user_interfaces.htm.